ECE 340: Semiconductor Devices

Lecture 22: Exam #1 Review

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

Office Hours: Thursdays 12p-1p, 2256 HMNTL

ECE 340 Midterm I

- Thursday October 15th
  - 7-8 pm, Arrive at Least 15 Minutes Early, Sit Every Other Seat
  - ECEB 1002, 1013, and 1015 – watch for messages on room assignments
- Conflicts
  - Conflict Defined Per UIUC Student Code
  - Must Contact TA Zetai Liu by Friday 10/9 at 5 pm
- DRES
  - Students Must Take Exams at TAC
  - Point of Contact Regardless of Section is John Dallesasse
  - Follow Instructions on Scheduling Provided Earlier
  - Must Provide Date/Time of Exam to TA Zetai Liu by Friday 10/9 at 5 pm
- Illness
  - See Policy in Syllabus, Excused Absence Letter from Dean's Office Required

Exam Location

Section A (Dallesasse): 1002 ECEB

Section B (Gilbert): 1002 ECEB

Section C (Kim): 1013 ECEB

Section D (He): 1015 ECEB

Thursday October 15th

7-8 pm (arrive early)

Major Topics to Date

- Crystal Structures, Density, and Miller Indices
- Energy Bands: Kinetic Energy, Potential Energy, E(k) Relationship and Effective Mass
- Donors, Acceptors, Electrons, and Holes
- Calculation of Carrier Concentration & Fermi Levels
- Charge Neutrality
- Drift Current, Resistance, and Conductance
  - Mobility
- Excess Carriers and Quasi Fermi Levels
- Thermal and Optical Generation & Recombination
- Diffusion and Diffusion Current

Important Equations

- Current Density Equations, Including Drift and Diffusion
- Charge Neutrality Equation, Fermi Function, Law of Mass Action
- Continuity Equation
- Diffusion Equation
- Einstein Relationship

Diamond Structure Bonding

Diamond Unit Cell (not primitive cell)

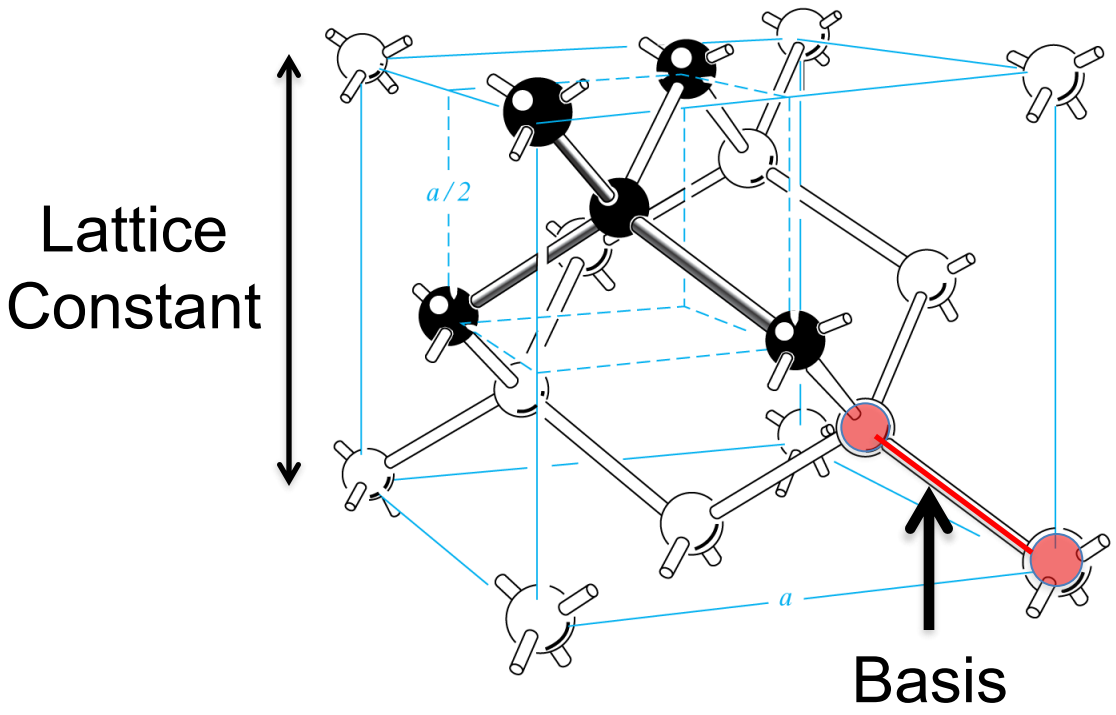


Figure 1.9

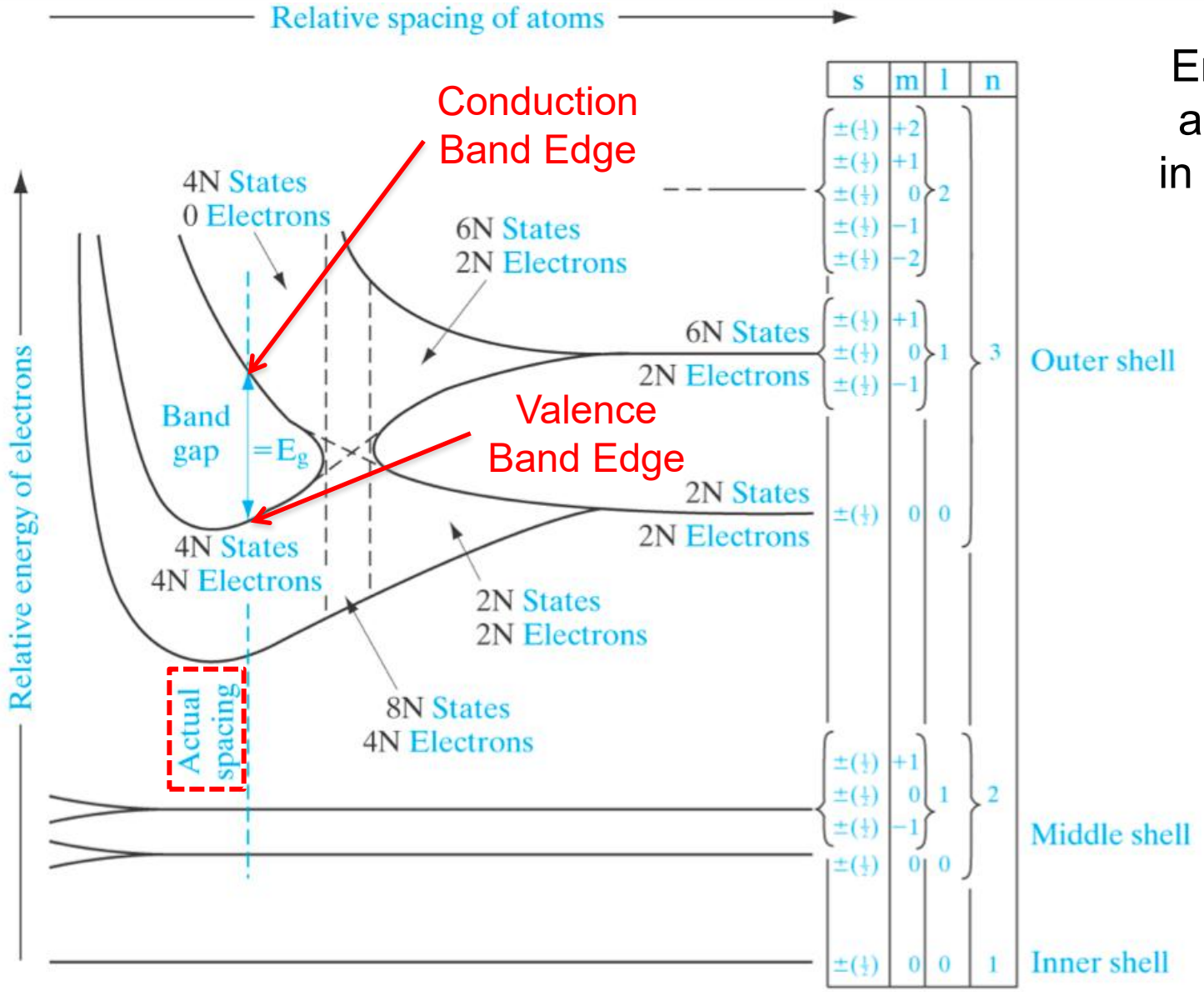
Diamond lattice unit cell, showing the four nearest neighbor structure.

(From *Electrons* and *Holes in Semiconductors* by W. Shockley, © 1950 by Litton Educational Publishing Co., Inc.; by permission of Van Nostrand Reinhold Co., Inc.)

- The **Diamond Structure** is an **FCC Lattice** with a **Basis** of 2 atoms
- The **Unit Cell** has **4 Lattice Points**, and **8 Atoms**

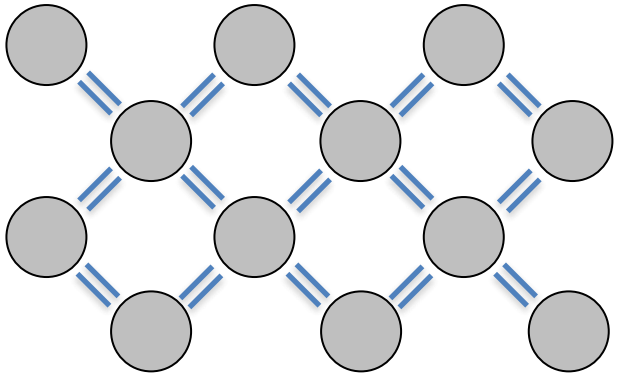
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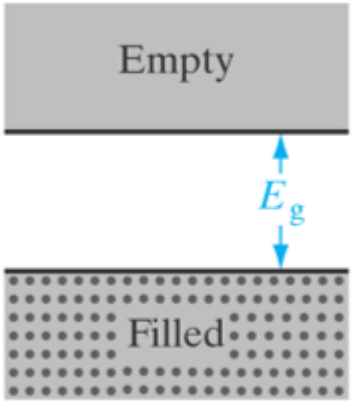
Formation of Energy Bands



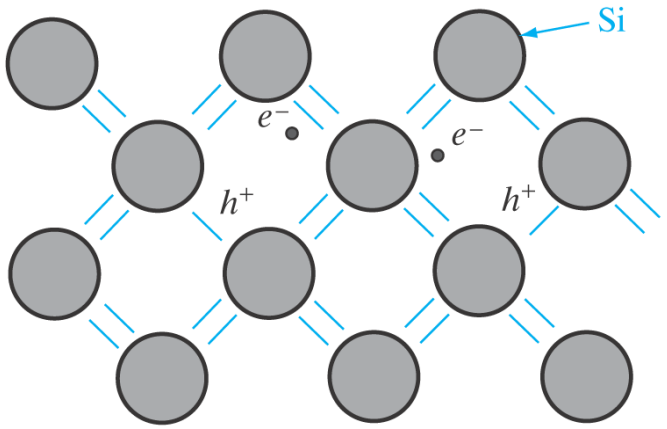
Energy bands and bandgap in silicon lattice

Electrons and Holes in "Perfect" Semiconductors





$n = p = n_i$



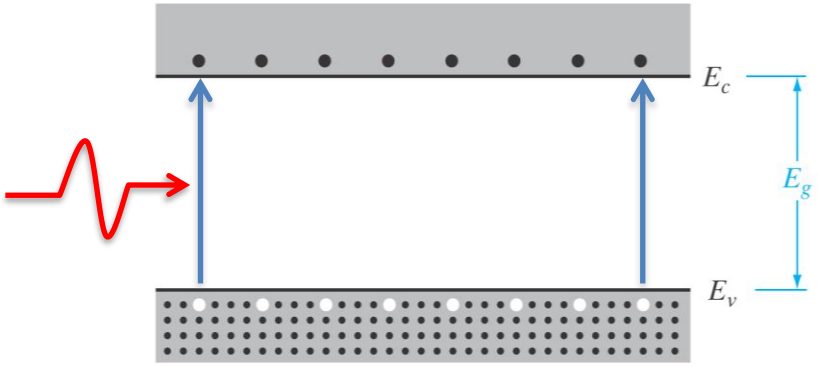


Figure 3.7

Electron–hole pairs in a semiconductor.

Electrons in Free Space

- As a consequence of wave-particle duality, the electron has a wave-like nature

$$\psi(z,t) = Ae^{[i(kz-\omega t)]}$$

where $k = \frac{2\pi}{\lambda}$ (wave vector); and $\omega = 2\pi\upsilon$ (frequency)

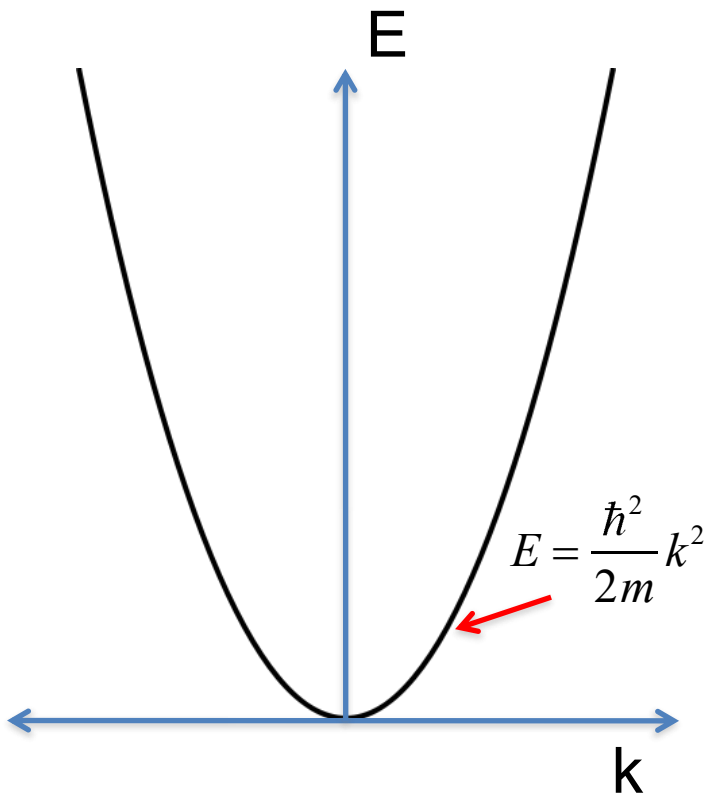
$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$ (note, here v is velocity)

but $p = \hbar k$

so $E = \frac{\hbar^2}{2m}k^2$

- The E(k) relationship is parabolic for an electron in free space
- The curvature at the origin is related to $\frac{\hbar^2}{2m}$

Classical Mechanics: $E = \frac{p^2}{2m}$



What would this look like if the mass were smaller? or negative?

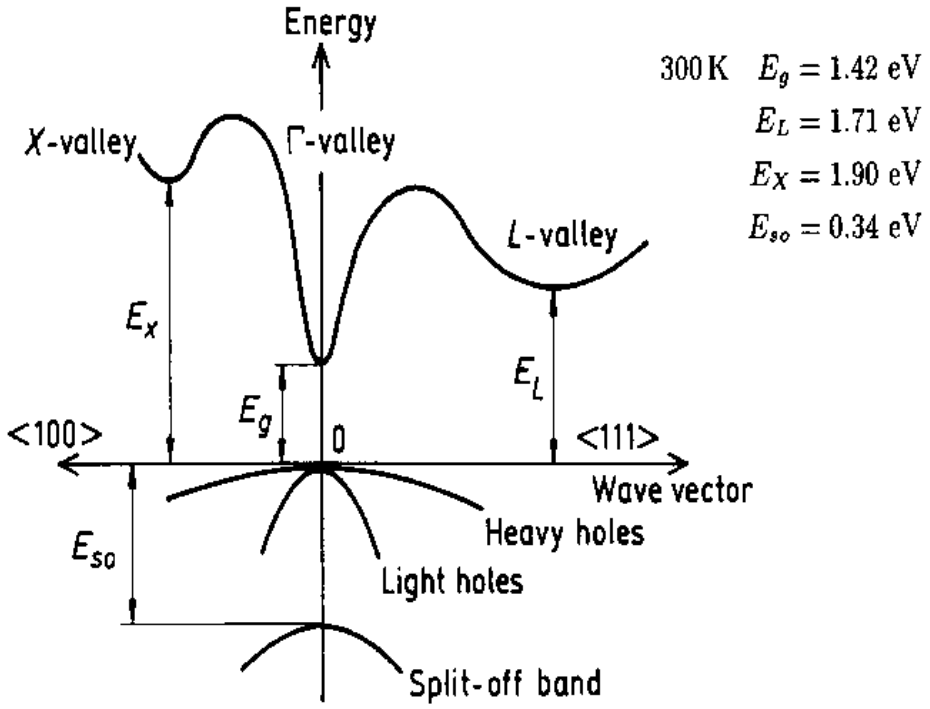
Electron and Hole Effective Mass

- In a semiconductor, the E(k) relationship looks parabolic near the conduction band minima and the valence band maxima
- **The curvature of the parabola differs from the free space curvature**, and depends upon which minima the electron or hole is in
- Near a minima, the E(k) relationship is:

$$E = \frac{\hbar^2}{2m^*}\mathbf{k}^2$$

- This is similar to the E(k) relationship for a free electron, but with a mass m* instead of m_o
- The parameter **m*** is the **effective mass**

GaAs Band Structure



http://www.ioffe.ru/SVA/NSM/Semicond/GaAs/bandstr.html

Electron in crystal lattice

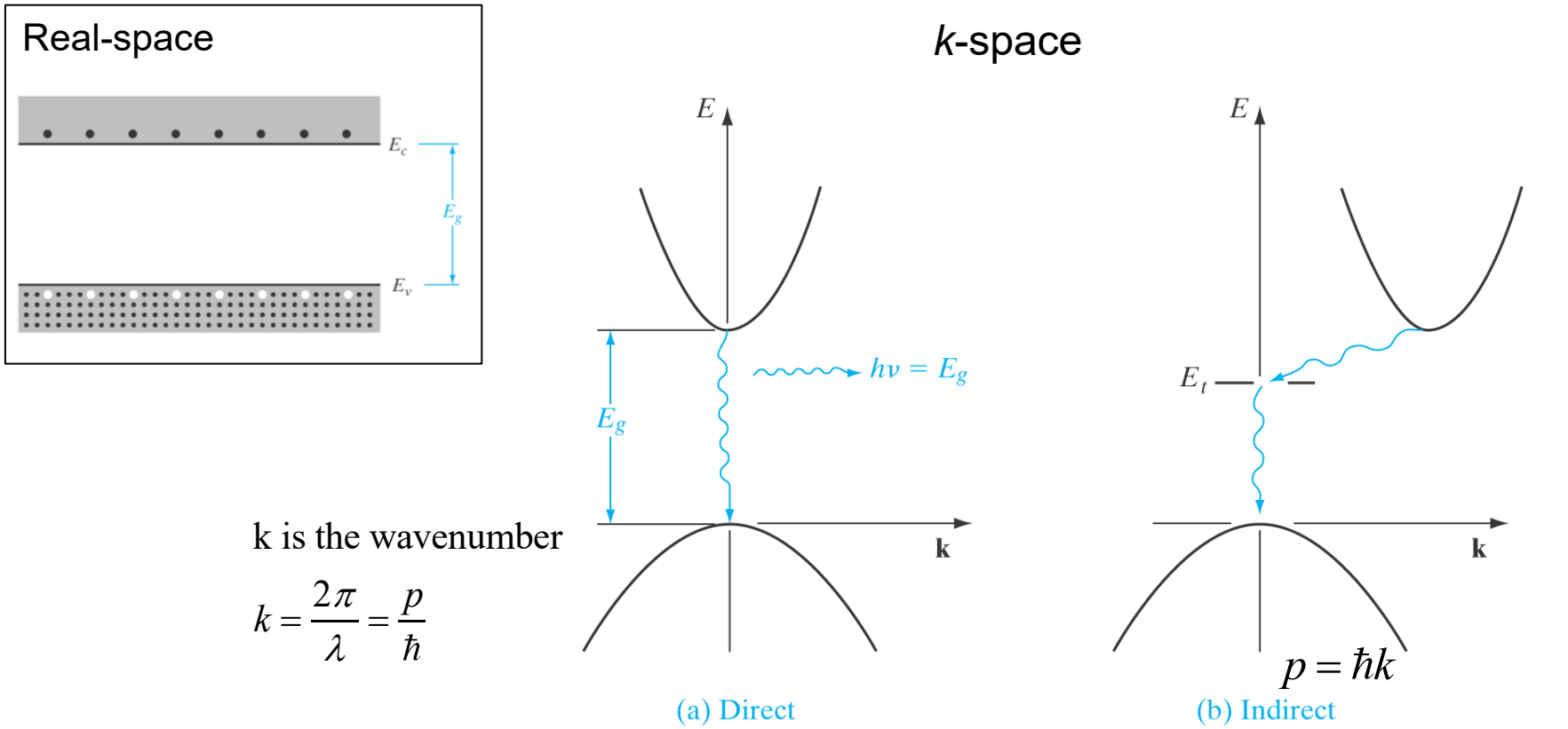


Figure 3.5

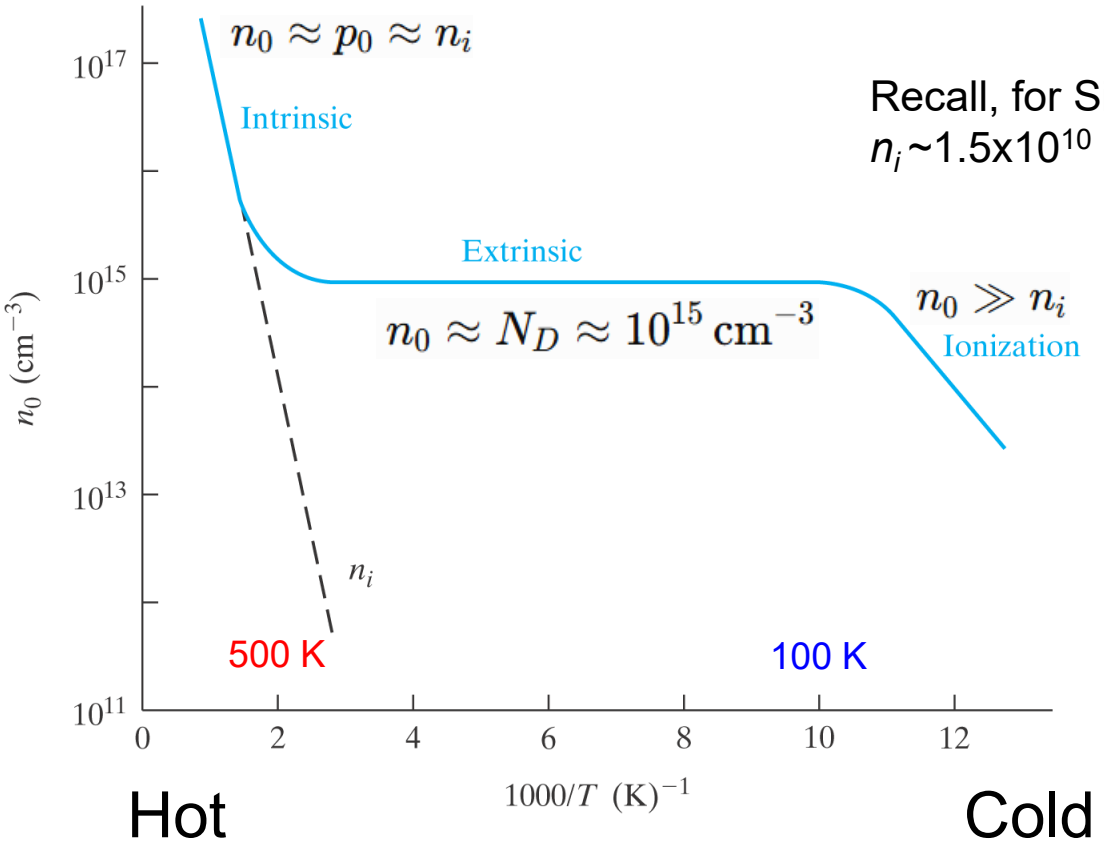
Direct and indirect electron transitions in semiconductors: (a) direct transition with accompanying photon emission; (b) indirect transition via a defect level.

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Extrinsic Material: Temperature Effects

Carrier concentration in Si doped with 10^{15} donors/cm^3

n_0: Electron concentration (in conduction band)



Recall, for Si, n_i ~1.5x10^{10} cm^{-3} at 300K

Figure 3.18

Carrier concentration vs. inverse temperature for Si doped with 10^{15} donors/cm^3.

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Fermi-Dirac Distribution

- The probability that an available energy state at E will be occupied by an electron at temperature T (temperature in Kelvin) is given by the **Fermi-Dirac Distribution Function**
- Fermi-Dirac distribution function:

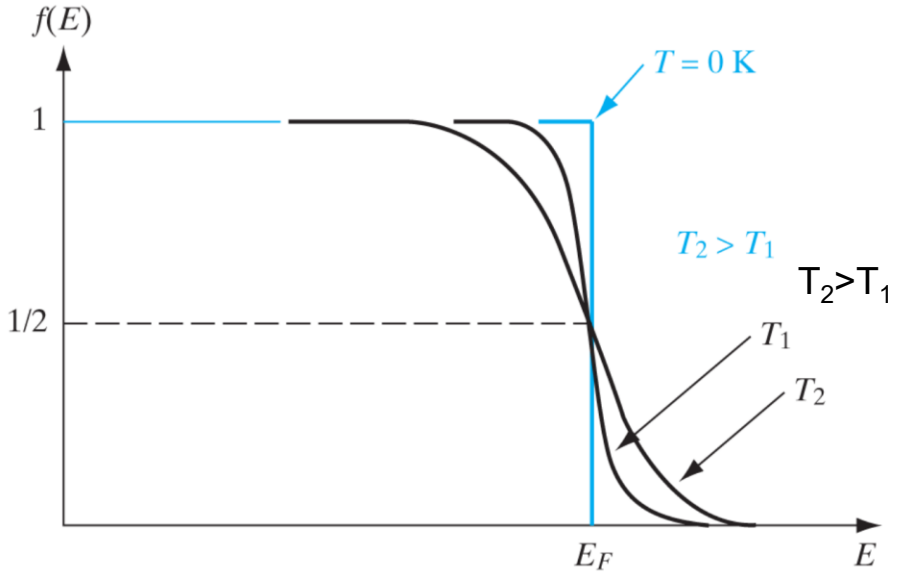
$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT}}$$

E_f : Fermi Level

k : Boltzmann Constant

$$k = 8.62 x 10^{-5}\, eV / K$$

- **Fermi level (E_f)**: the energy that determines how likely an electronic states is to be occupied by an electron.
- The Fermi level has units of energy (typically eV)



(a) $E = E_f$ $\quad f(E = E_f) = \frac{1}{1 + e^0} = \frac{1}{2}$

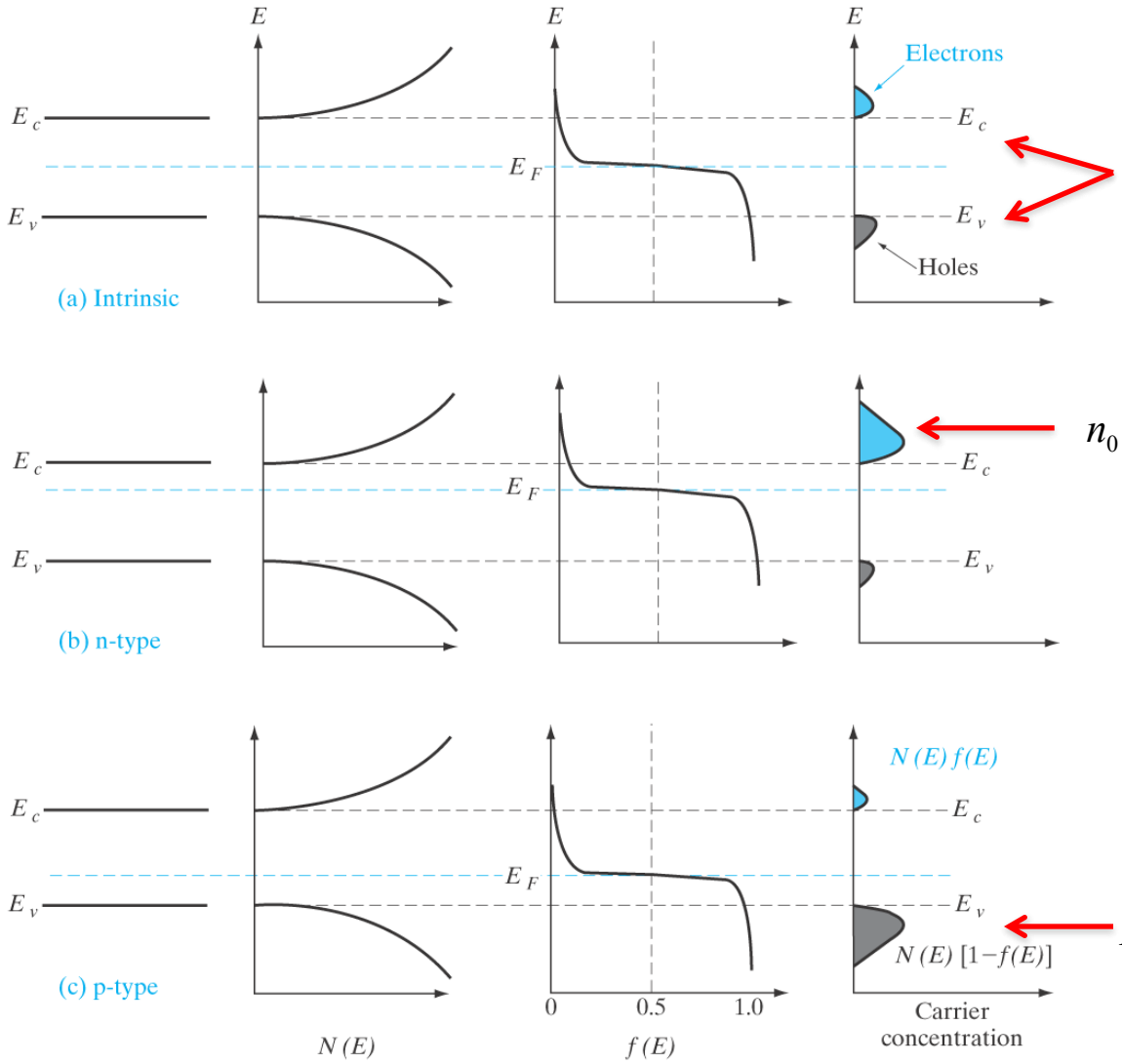
(b) T = 0K

$$f(E < E_f) = \frac{1}{1 + e^{-\infty}} = 1$$

$$f(E > E_f) = \frac{1}{1 + e^{+\infty}} = 0$$

Carrier Concentration at Thermal Equilibrium

The Fermi Function Multiplied by the Density of States



n_i, p_i: intrinsic concentration

Conduction band DOS

$$n_i = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= p_i = \int_{-\infty}^{E_V} [1 - f(E)] N_V(E) dE$$

Probability the state is empty

Valence band DOS

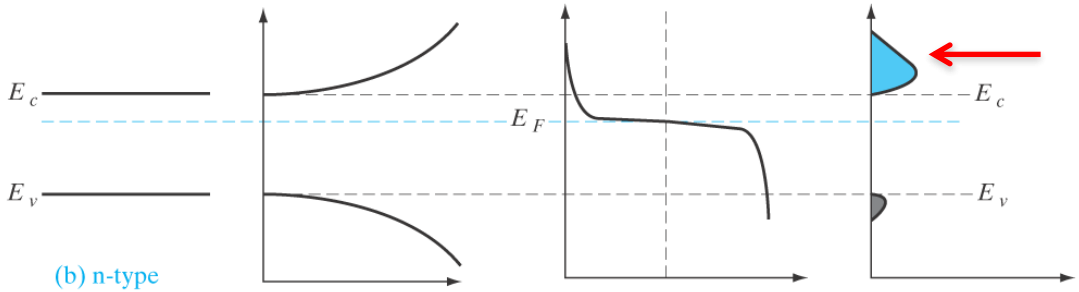
$$n_0 = \int_{E_c}^{\times} f(E) N_c(E) dE$$

n_0, p_0: equilibrium concentration (could be intrinsic or extrinsic)

Thus, $n_0 = p_0 = n_i$ for intrinsic Si

$$p_0 = \int_{-\infty}^{E_v} [1 - f(E)] N_v(E) dE$$

Carrier Concentration n_o at Equilibrium



$$n_0 = \int_{E_c}^{\times} f(E) N_c(E) dE$$

$$= \int_{E_c}^{\infty} \frac{1}{1+e^{(E-E_f)/kT}} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE$$

$$f(E_c) = e^{-(E_c-E_f)/kT} \qquad if \qquad e^{(E_c-E_f)/kT} >> 1$$

$$n_o = \int_{E_c}^{\infty} e^{-(E-E_f)/kT} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2} e^{(E_F-E_C)/kT}$$

$$= N_C(T) e^{-(E_C-E_F)/kT}$$

$$n_o = N_c \bullet e^{-(E_c-E_f)/kT}$$

where $$N_c(T) = 2 \left(\frac{2\pi \bullet m_n^* kT}{h^2} \right)^{3/2}$$

Similarly, $$p_o = N_v \bullet e^{-(E_f-E_v)/kT}$$

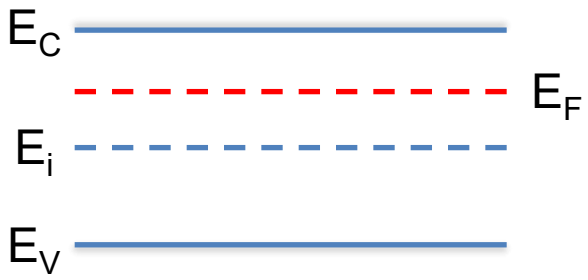
where $$N_v(T) = 2 \left(\frac{2\pi \bullet m_p^* kT}{h^2} \right)^{3/2}$$

Complete solution : Appendix IV

Non-Degenerate Case: Concentration Product and Intrinsic Concentration

Non-degenerate: E_F far enough from E_C or E_V (>3kT) for the approximation to be valid.

$$f(E_C) = \frac{1}{1+e^{(E_C-E_f)/kT}} \sim e^{-(E_C-E_f)/kT}$$



$$n_i p_i = N_c e^{-(E_c-E_i)/kT} \square N_v e^{-(E_i-E_v)/kT} = N_c N_v e^{-(E_c-E_v)/kT} = N_c N_v e^{-E_g/kT}$$

$$n_o p_o = N_c e^{-(E_c-E_f)/kT} \square N_v e^{-(E_f-E_v)/kT} = N_c N_v e^{-(E_c-E_v)/kT} = N_c N_v e^{-E_g/kT}$$

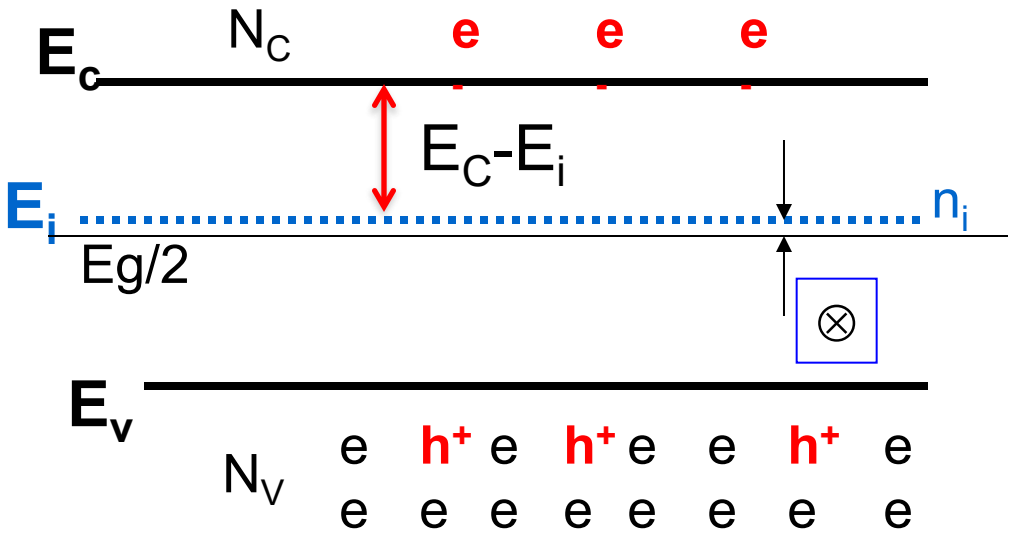
therefore $n_o p_o = n_i p_i = n_i^2$

and $n_i^2(T) = N_c N_v \square e^{-(E_g)/kT}$

$$n_i(T) = \sqrt{N_c N_v} \square e^{-(E_g)/2kT}$$

- The intrinsic carrier concentration depends upon
  - Bandgap, temperature, electron effective mass (through N_c), and hole effective mass (through N_v)
- E_i is at midgap only if $N_c = N_v$ (equal electron and hole effective masses)

Intrinsic Fermi Level: Not Quite Midgap



$$n_i(T) = \sqrt{n_i p_i} = \sqrt{N_c N_v}\,e^{-(E_g)/2kT}$$

and $$n_i(T) = N_c\,e^{-(E_c - E_i)/kT}$$

so

$$N_c\,e^{-(E_c - E_i)/kT} = \sqrt{N_c N_v}\,e^{-(E_g)/2kT}$$

$$e^{(E_g)/2kT}\,e^{-(E_c - E_i)/kT} = \left\{\frac{N_v}{N_c}\right\}^{1/2}$$

$$\frac{E_g}{2} - (E_c - E_i) = \Delta = kT \ln\left\{\frac{N_v}{N_c}\right\}^{1/2}$$

$$\Delta = kT \ln\left\{\left(\frac{m_p^*}{m_n^*}\right)^{3/2}\right\}^{1/2}$$

If Δ is positive, E_i is closer to E_c than midgap.

If Δ is negative, E_i is closer to E_v than midgap.

Δ is positive when $m_p^* > m_n^*$

Charge Neutrality

Charge Neutrality Equation :

$$p_o + N_d^+ = n_o + N_a^-$$

N_d^+ : ionized donor (positively charged)
N_a^- : ionized acceptor (negatively charged)

Example Revisited:

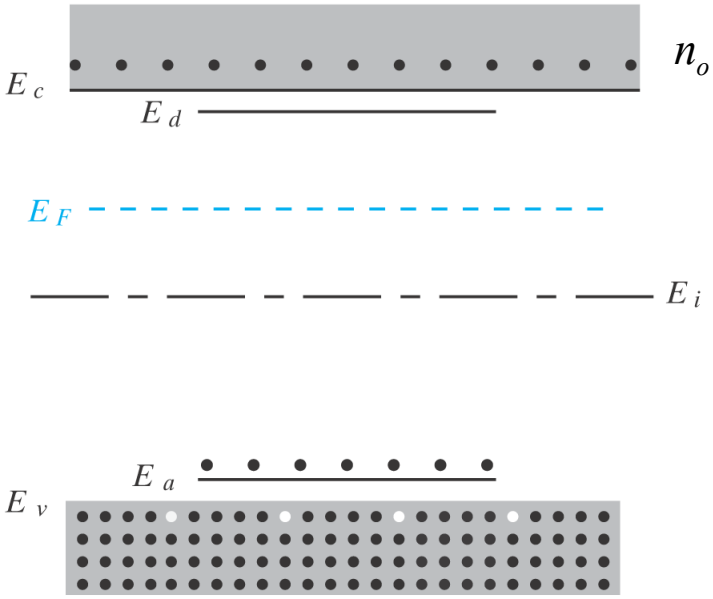
Donor-Doped Silicon, 10^{17} cm^{-3}

Since $N_d(As) >> n_i$,

$$n_o \approx N_d = 10^{17}\,cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(1.5 \times 10^{10}\,cm^{-3}\right)^2}{10^{17}\,cm^{-3}}$$

$$p_o = 2.25 \times 10^{3}\,cm^{-3} << 10^{17}\,cm^{-3}$$



$$n_o = \left(N_d^+ - N_a^-\right) + p_o \approx N_d^+ - N_a^-$$

Figure 3.19

Compensation in an n-type semiconductor ($N_d > N_a$).

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Drift Current in a Constant Field

Electron Current Relationships

$$J_{x,n} = -qn\langle v_x \rangle = qn\mu_n E_x = \sigma_n E_x$$

Electron Conductivity: $\sigma_n = qn\mu_n$

Electron Mobility: $\mu_n = \frac{q\bar{t}_n}{m_n^*}$

Hole Current Relationships

$$J_{x,p} = qp\langle v_x \rangle = qp\mu_p E_x = \sigma_p E_x$$

Hole Conductivity: $\sigma_p = qp\mu_p$

Hole Mobility: $\mu_p = \frac{q\bar{t}_p}{m_p^*}$

Combined Current

$$J_{x,total} = q(n\mu_n + p\mu_p)E_x = \sigma E_x$$

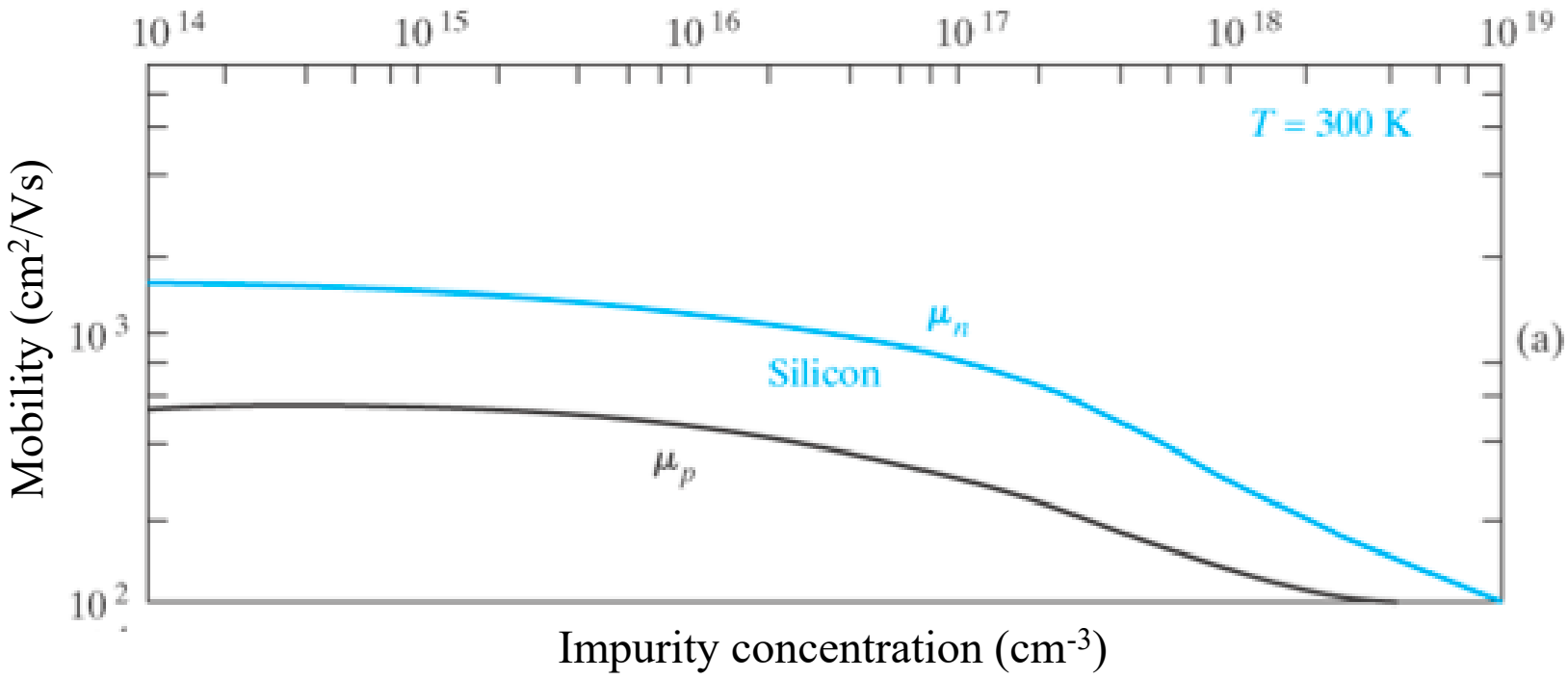
$$\sigma = q(n\mu_n + p\mu_p)$$

$V = IR$, or $I = \frac{1}{R}V$

We've made it to V=IR!

What Value Do I Use for Mobility?

$$\mu \sim \frac{q\bar{t}}{m^*}$$



Impurity Concentration, Not Carrier Concentration

N_D+N_A

Temperature effects

Mobilities Add Inversely:

$$\frac{1}{\mu_{tot}} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \ldots$$

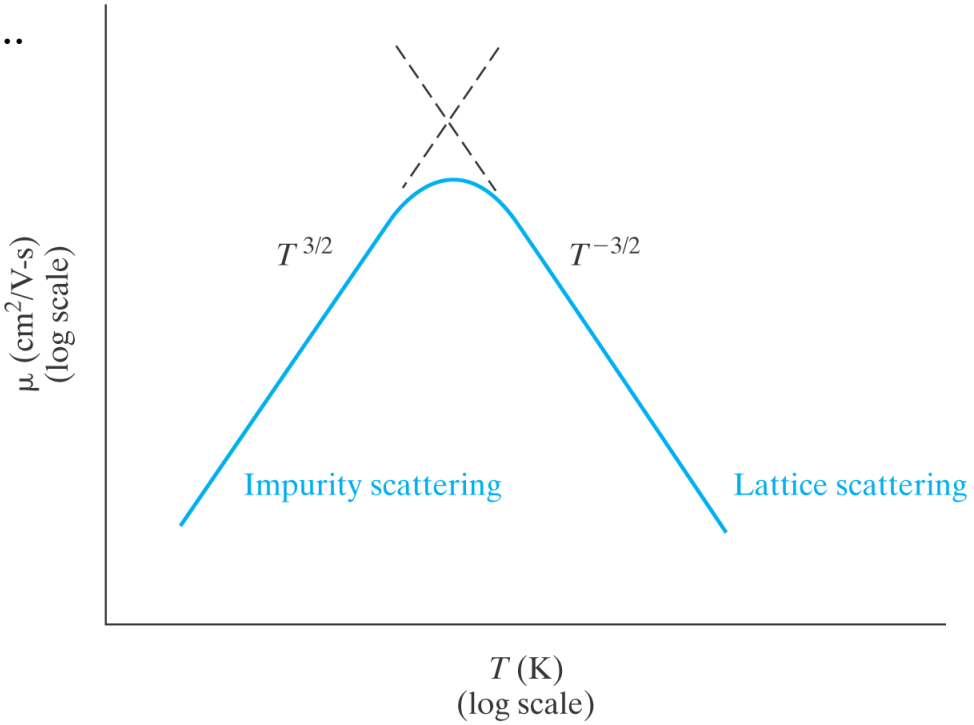


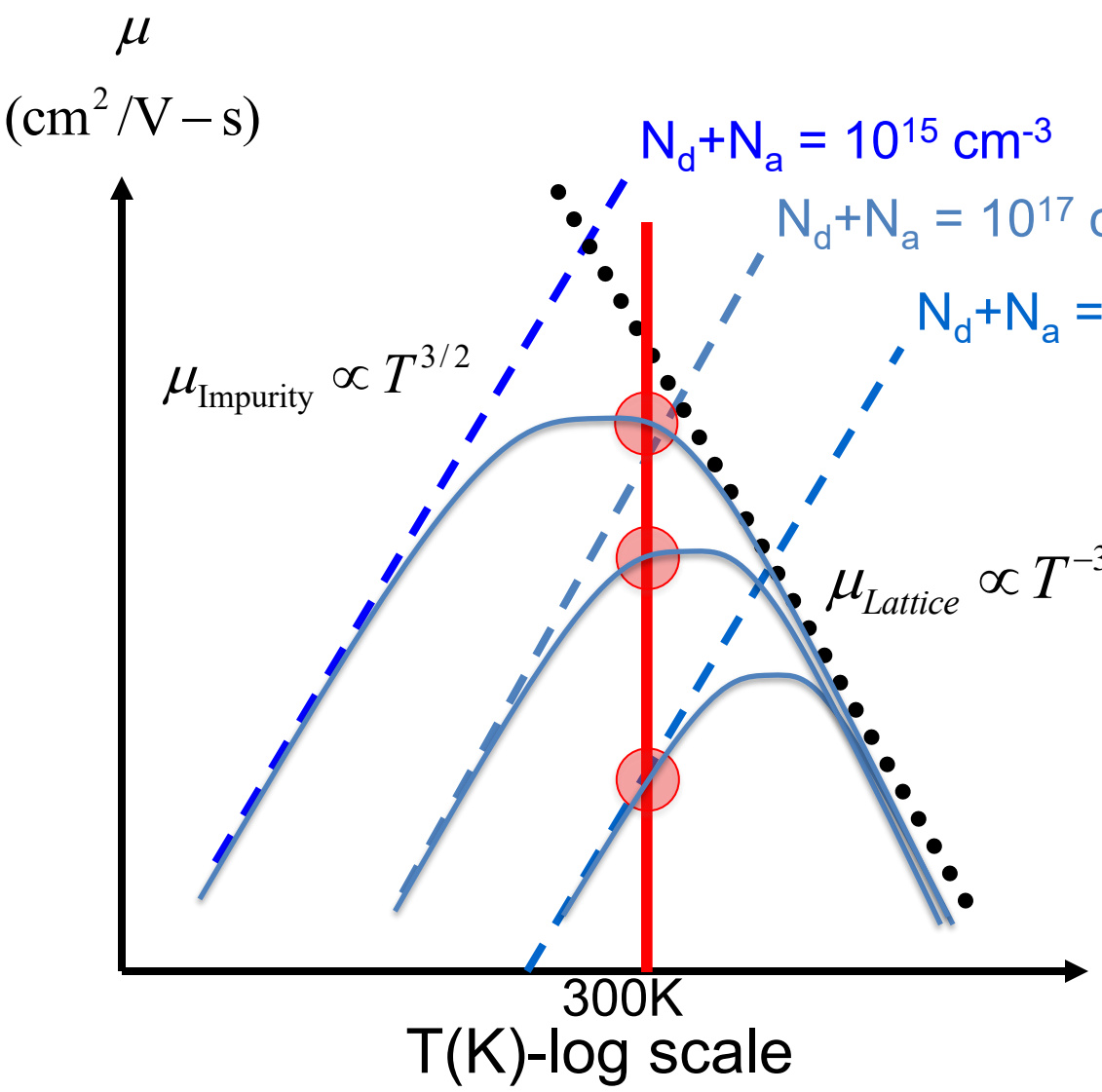
Figure 3.22

Approximate temperature dependence of mobility with both lattice and impurity scattering.

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Effects of Both Temperature and Impurity Concentration on Mobility

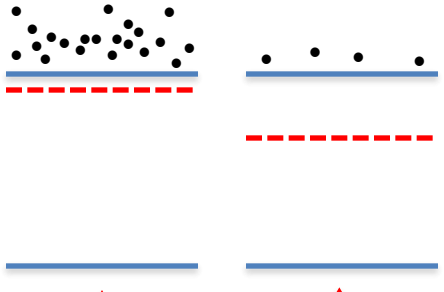
$$\frac{1}{\mu} = \frac{1}{\mu_{Impurity}} + \frac{1}{\mu_{Lattice}} + \frac{1}{\mu_{Defect}} + \ldots$$



At T=300K

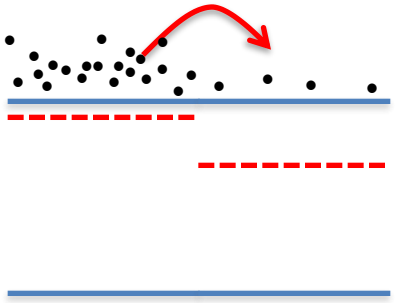
- For $N_d+N_a=10^{19}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{17}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{15}$cm^{-3}, the mobility is dominated by lattice scattering

Establishing Equilibrium

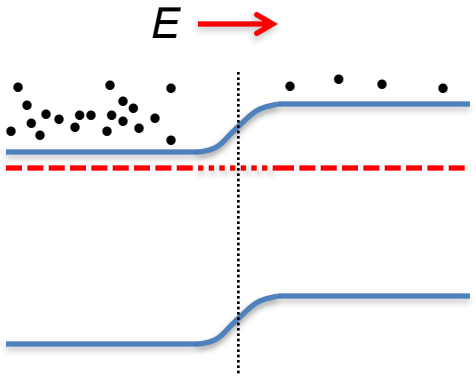


High Carrier Concentration

Low Carrier Concentration



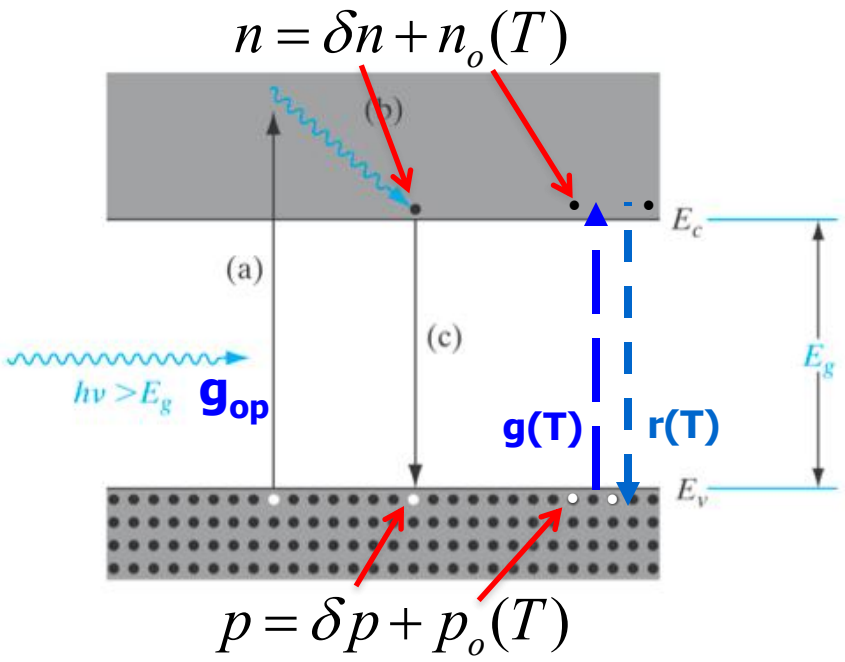
Carrier Diffusion (pre equilibrium)



Equilibrium: Diffusion Balanced by Drift (Electric Field)

We Will Come Back to this in the Discussion of pn Junctions!

Optical Absorption of Photons and Excess Carriers

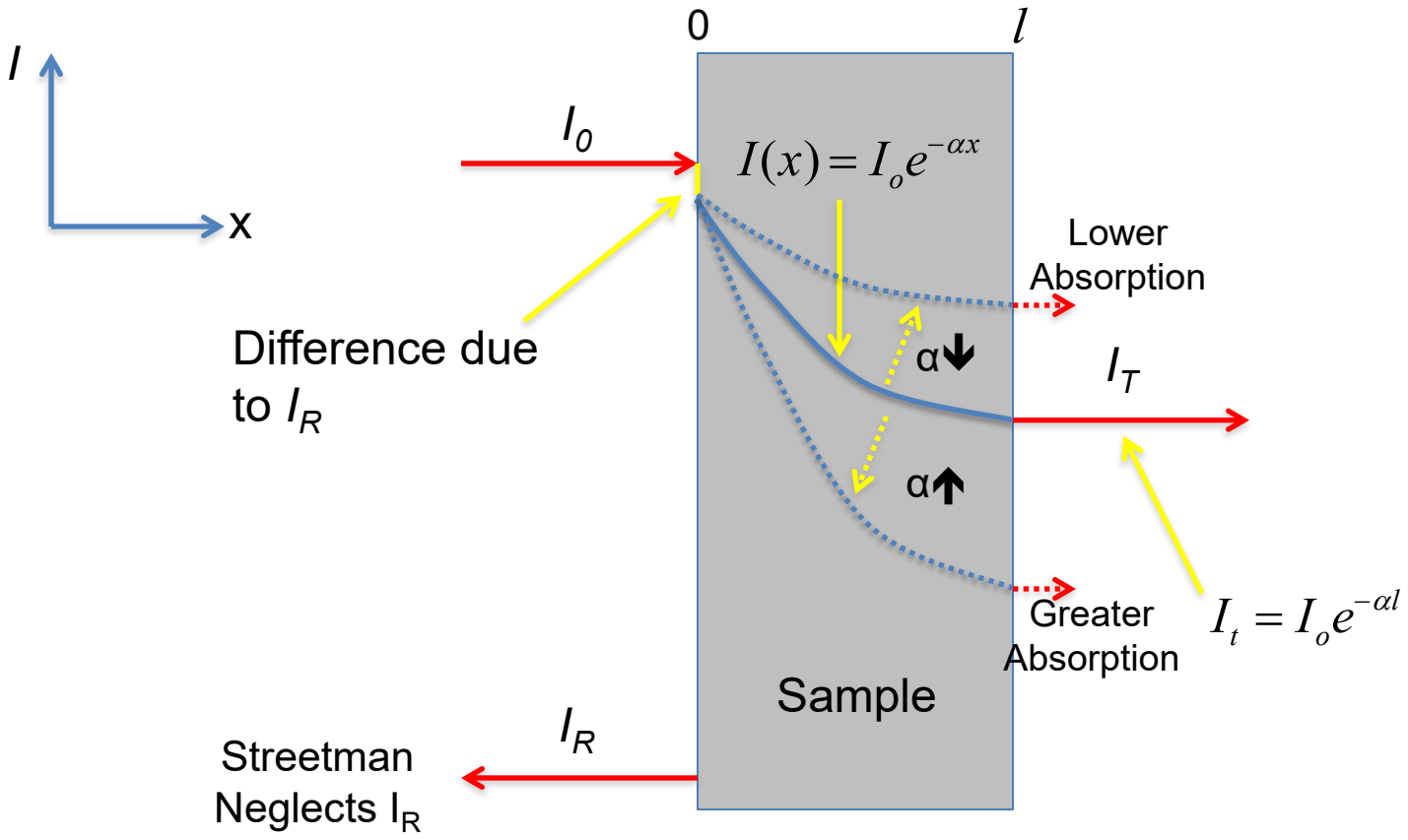


- A photon with energy $hv<E_g$ will not be absorbed. Hence, it is transmitted through the sample.
  - Insulators are frequently transparent due to their large bandgap
- A photon with energy $hv>E_g$ can be absorbed to excite an electron from the valance band to the conduction band.
- Thermal generation of carriers provides the equilibrium carrier concentrations, $n_o(T)$ and $p_o(T)$
- Optical generation of excess carriers (δn and δp) can contribute to the conductivity

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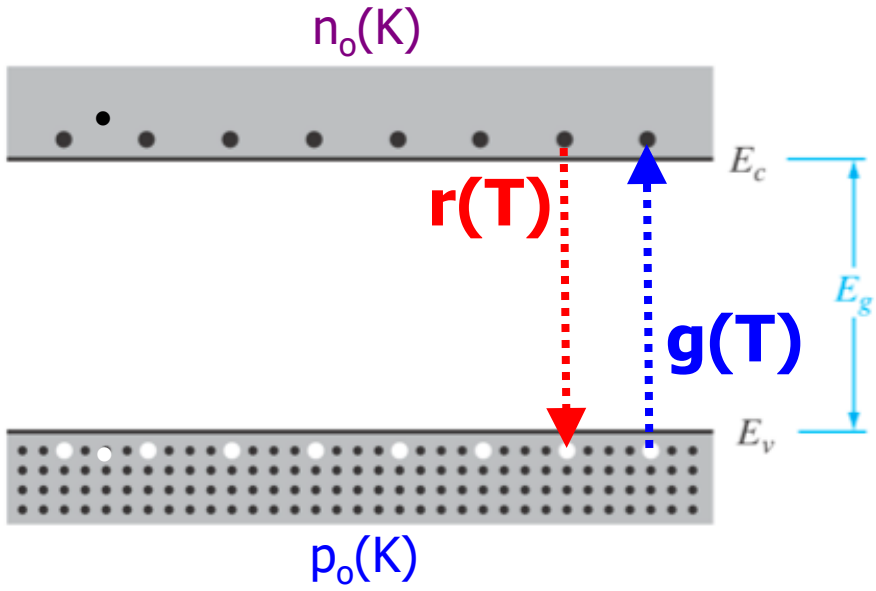
Photon Flux Versus Position

- The photon flux drops exponentially in the sample
- The relationship for the generated carriers is more complicated due to diffusion



Equilibrium Generation

Some electrons have sufficient energy to jump to the conduction band (phonon interactions)



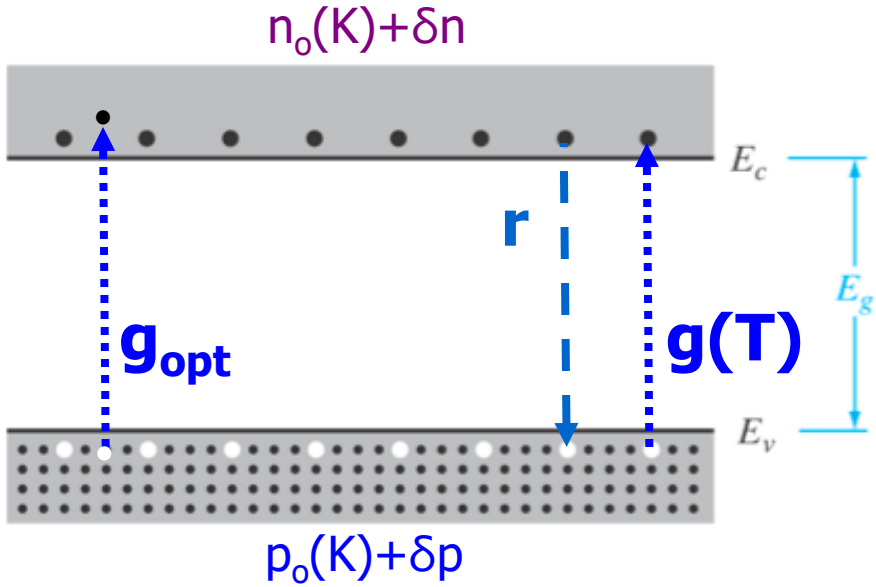
Generation and recombination are balanced:

$$g(T) = g_i = r_i = \alpha_r n_o p_o = \alpha_r n_i^2$$

α_r : e-h recombination coefficient
(Not the same as the absorption coefficient α)

Steady State Generation

$$g = g(T) + g_{opt} = r = \alpha_r np$$
$$= \alpha_r (n_0 + \delta n)(p_0 + \delta p)$$
$$= \alpha_r \left[n_0 p_0 + n_0 \delta p + p_0 \delta n + \delta n \delta p \right]$$
$$= \alpha_r \left[n_0 p_0 + (n_0 + p_0)\delta n + \delta n^2 \right]$$
$$= \alpha_r n_0 p_0 + \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right]$$



Recall: $g(T) = \alpha_r n_i^2 = \alpha_r n_0 p_0$

so

$$g_{op} = \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right] \approx \alpha_r (n_0 + p_0)\delta n = \frac{\delta n}{\tau_n}$$

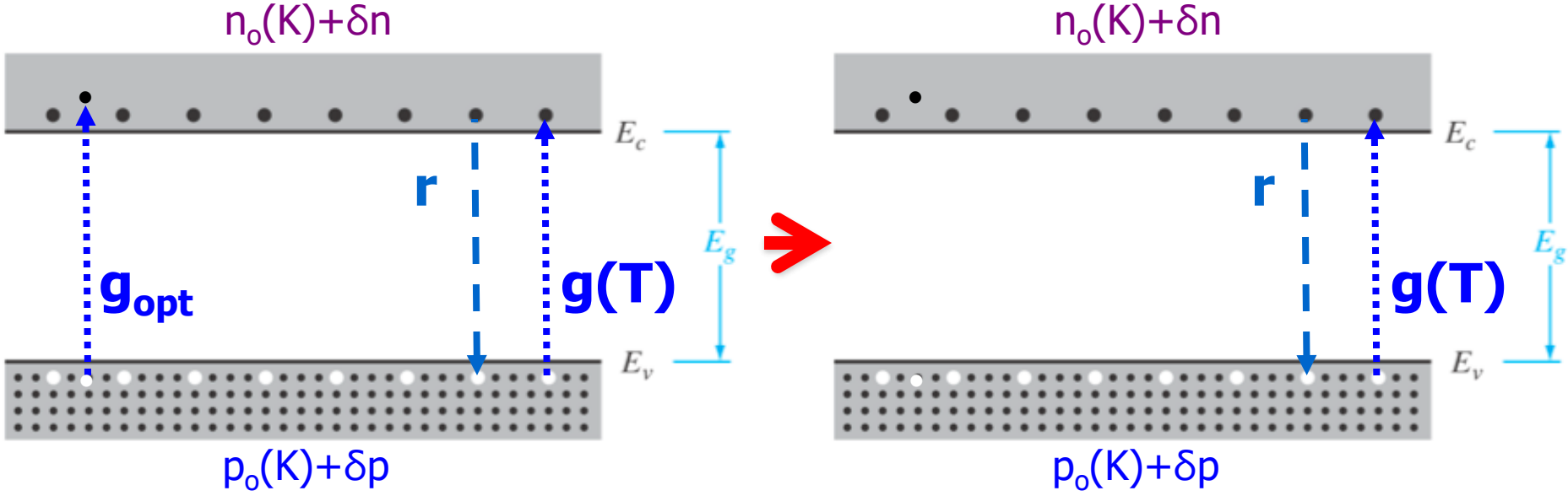
$$\tau_n = \frac{1}{\alpha_r (n_0 + p_0)}$$

and

$$\delta n = \delta p = g_{op} \tau_n$$

Assume *δn* is small compared to majority carrier concentration

Optical Generation Stops: $g_{opt}(t=0)=0$



Recombination: $r = g(T) + g_{opt}$

Generation: $g(T) + g_{opt}$

Generation and recombination are balanced

Recombination: $r = g(T) + g_{opt}$

Generation: $g(T)$

Generation and recombination are not balanced

The number of excess carriers decreases:

$$\frac{dn(t)}{dt} = g(T) - r(t) = \alpha_r n_i^2 - \alpha_r n(t) p(t)$$

Solving the Differential Equation

p-type Material

1) Assume low-level injection:

neglect $\delta n^2(t)$

2) p-type material so $p_0 >> n_0$

$$\frac{d\delta n(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta n(t) + \delta n^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta n(t) \approx -\alpha_r p_0 \delta n(t)$$

The solution is:

$$\delta n(t) = \Delta n e^{-\alpha_r p_0 t} = \Delta n e^{-t/\tau_n}$$

where

$$\Delta n = \delta n(t = 0)$$

$$\tau_n = \frac{1}{\alpha_r p_0} \quad \text{or}$$

$$\tau_n = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

n-type Material

1) Assume low-level injection:

neglect $\delta p^2(t)$

2) n-type material so $n_0 >> p_0$

$$\frac{d\delta p(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta p(t) + \delta p^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta p(t) \approx -\alpha_r n_0 \delta p(t)$$

The solution is:

$$\delta p(t) = \Delta p e^{-\alpha_r n_0 t} = \Delta p e^{-t/\tau_p}$$

where

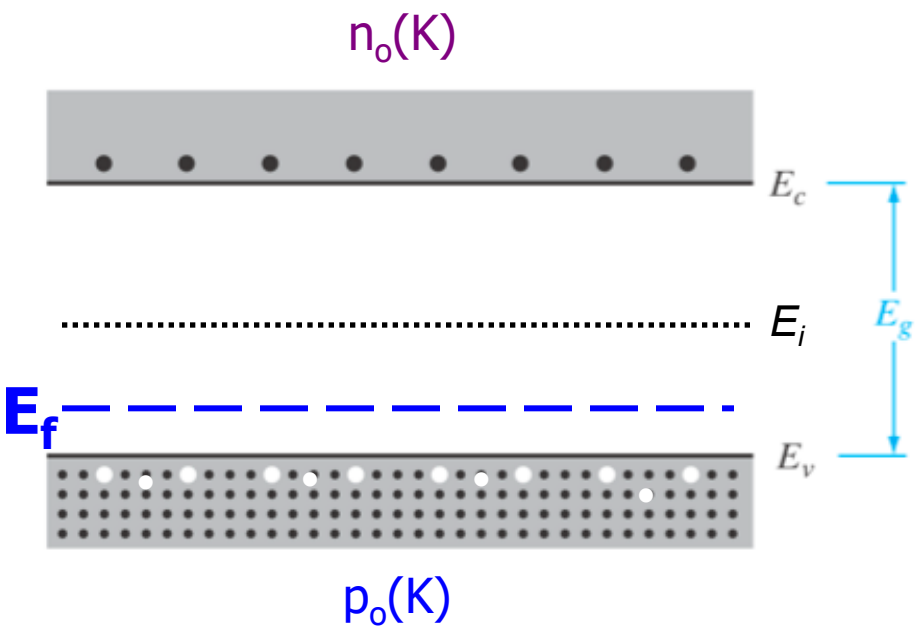
$$\Delta p = \delta p(t = 0)$$

$$\tau_p = \frac{1}{\alpha_r n_0} \quad \text{or}$$

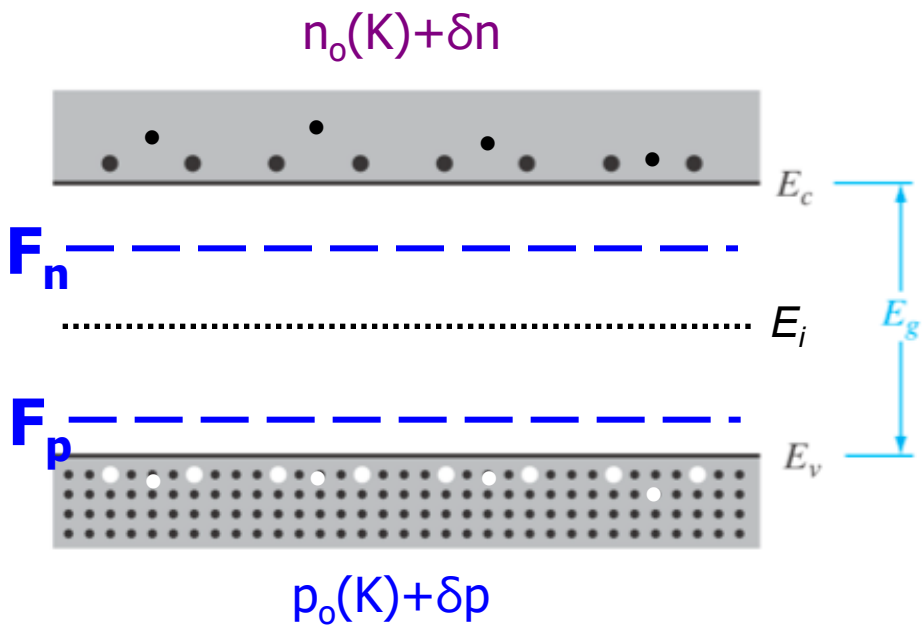
$$\tau_p = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

Non-equilibrium Fermi level

Equilibrium



With Excess Carriers



The Fermi level in the presence of excess carriers is called the "quasi-Fermi level"

$$n_0 = n_i e^{(E_F - E_i)/kT}$$

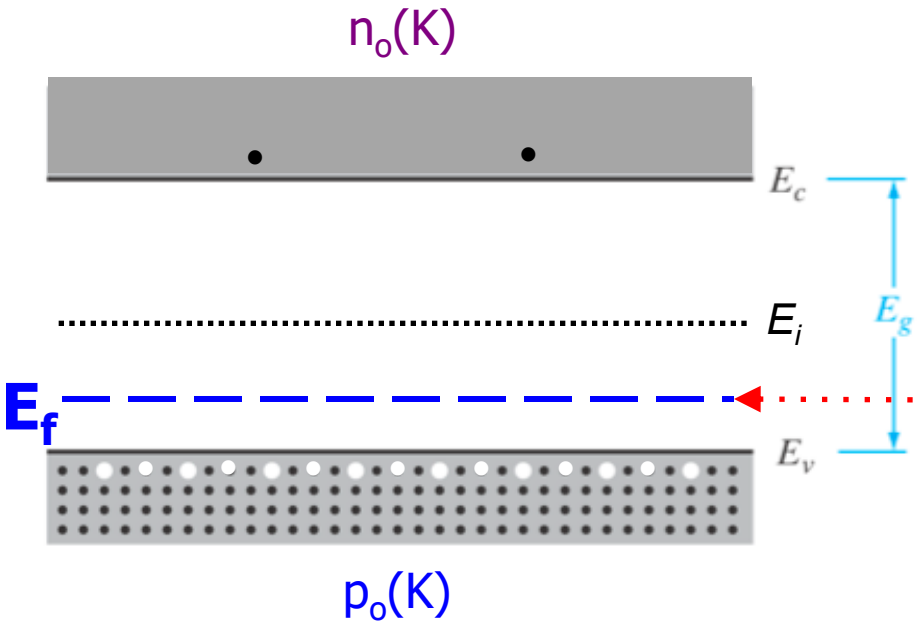
$$p_0 = n_i e^{(E_i - E_F)/kT}$$

$$n = n_i e^{(F_n - E_i)/kT}$$

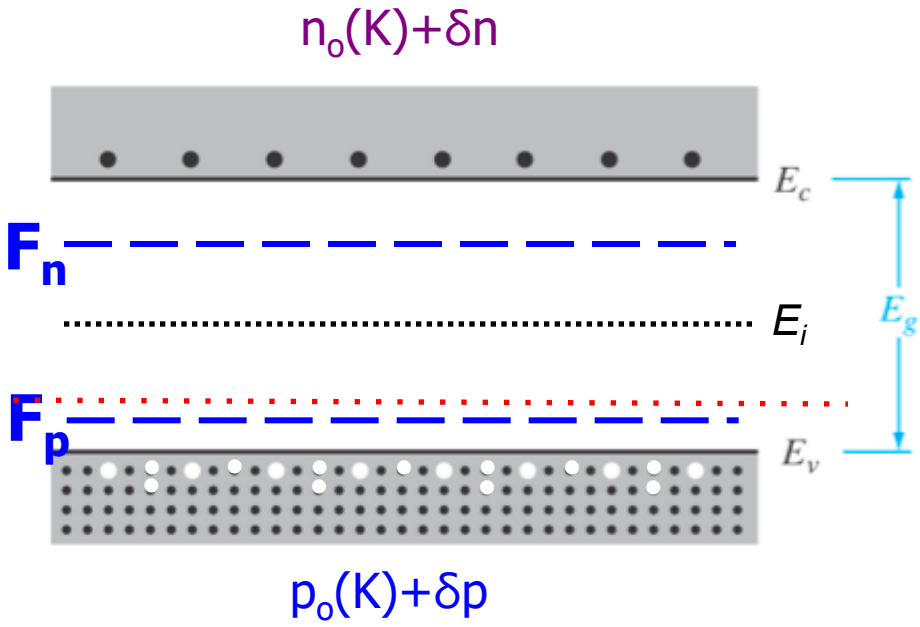
$$p = n_i e^{(E_i - F_p)/kT}$$

Quasi-Fermi Levels in p-type Material

Equilibrium



With Excess Carriers



$$n = n_i e^{(F_n - E_i)/kT}$$

$$F_n - E_i = kT \ln\left(\frac{n}{n_i}\right)$$

Diffusion Current

$$\phi_n(x_o) = \frac{\bar{l}}{2\bar{t}}(n_1 - n_2)$$

Since $\bar{l}$ is small: $n(x) - n(x+\Delta x) = (n_1 - n_2)\frac{\Delta x}{\bar{l}}$ and thus $(n_1 - n_2) = \frac{n(x) - n(x+\Delta x)}{\Delta x}\bar{l}$

In the limit of small Δx: $\phi_n(x) = \frac{\bar{l}}{2\bar{t}} \lim_{\Delta x \to 0}\left(\frac{n(x) - n(x+\Delta x)}{\Delta x}\right) = \frac{-\bar{l^2}}{2\bar{t}}\frac{dn(x)}{dx}$

The term $\frac{\bar{l^2}}{2\bar{t}}$ is the electron diffusion coefficient D_n with units cm^2/s

$$D_n = \frac{\bar{l^2}}{2\bar{t}}$$

A similar relation can be derived for hole motion. So:

$$\phi_n(x) = -D_n\frac{dn(x)}{dx}$$

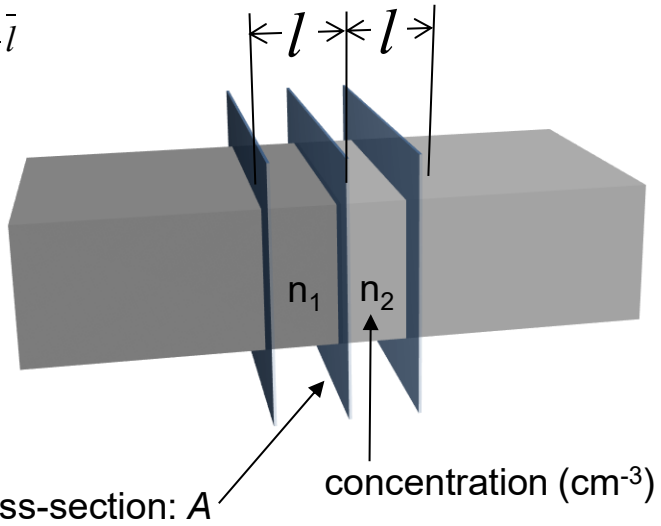
$$\phi_p(x) = -D_p\frac{dp(x)}{dx}$$

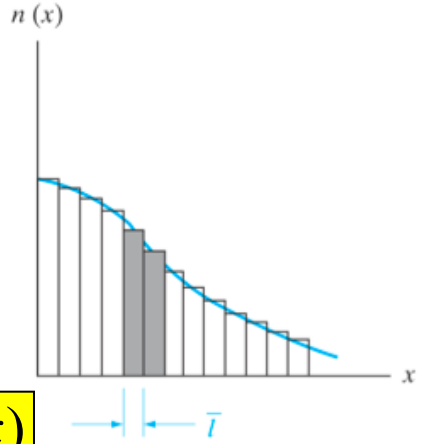
Electron and hole flux can be expressed as current:

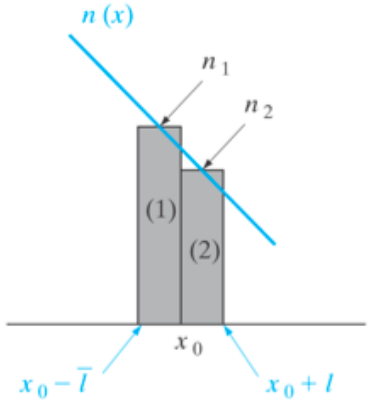
$$J_n(diff.) = -D_n\frac{dn(x)}{dx} \times (-q) = qD_n\frac{dn(x)}{dx}$$

$$J_p(diff.) = -D_p\frac{dp(x)}{dx} \times (+q) = -qD_p\frac{dp(x)}{dx}$$

$$J_n(x) = qD_n\frac{dn(x)}{dx} \text{ and } J_p(x) = -qD_p\frac{dp(x)}{dx}$$







Total Current: Drift and Diffusion

$$J_n(x) = J_n(drift) + J_n(diff)$$

$$\rightarrow J_n(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx}$$

$$J_p(x) = J_p(drift) + J_p(diff)$$

$$\rightarrow J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

$$\rightarrow J(x) = J_n(x) + J_p(x)$$

$$J(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx} + qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

Derivation of Einstein Relationship

$$J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx} = 0$$

$$\text{so } \mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{p(x)}\frac{dp(x)}{dx}$$

$$p_0(x) = n_i e^{(E_i(x)-E_F(x))/kT}.$$

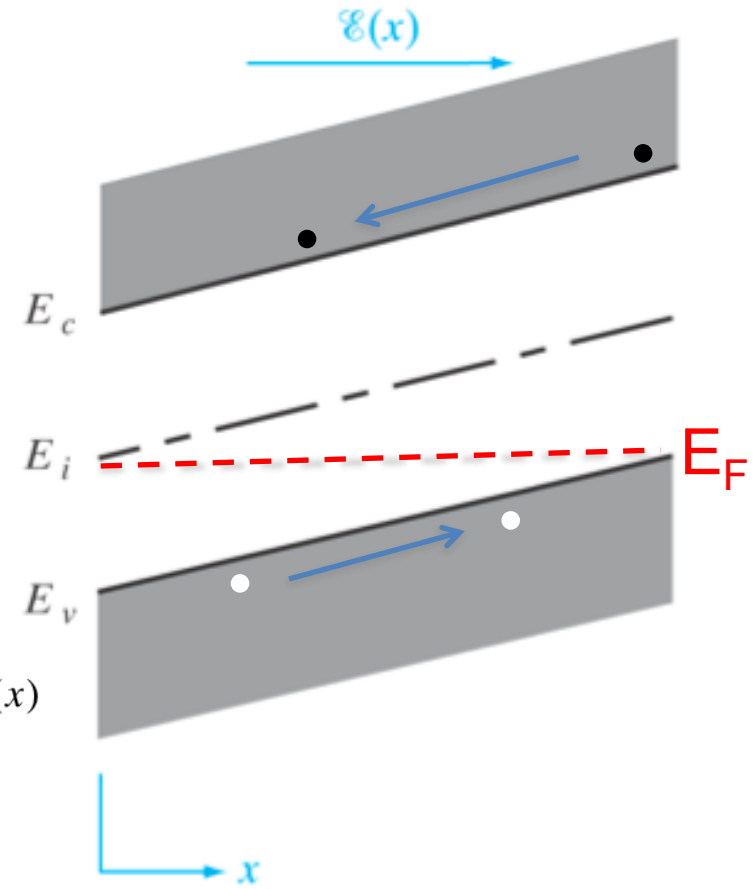
$$\text{using } \frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

$$\frac{dp_0}{dx} = n_i e^{(E_i-E_F)/kT}\frac{d}{dx}\left[\frac{E_i - E_F}{kT}\right]$$

$$\mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right) = \frac{D_p}{\mu_p}\frac{1}{kT}(q\mathcal{E}(x) - 0) \text{ since } \frac{1}{q}\frac{dE_i}{dx} = \mathcal{E}(x)$$

$$\text{so } 1 = \frac{D_p}{\mu_p}\frac{q}{kT} \text{ and } \boxed{\frac{D_p}{\mu_p} = \frac{kT}{q}}$$

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

Continuity Equations

$$\frac{\partial n}{\partial t} = G_n - U_n + \frac{1}{q}\nabla \bullet \vec{J}_n$$

$$\frac{\partial p}{\partial t} = G_p - U_p - \frac{1}{q}\nabla \bullet \vec{J}_p$$

Divergence of a vector field

For a given volume of semiconductor, the rate change of carrier is the net effect of current flow into the volume and generation and recombination rates within the volume.

G_n : *electron generation rate*

G_p : *hole generation rate*

U_n : *electron recombination rate*

U_p : *hole recombination rate*

Diffusion Equation Where Drift is Negligible

Hole Diffusion Equation

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$$

$$\text{and } J_p(x) = -qD_p\frac{dp(x)}{dx} = -qD_p\frac{\partial \delta p}{\partial x} \text{ (diffusion)}$$

$$\text{so } \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial}{\partial x}\left(-qD_p\frac{\partial \delta p}{\partial x}\right) - \frac{\delta p}{\tau_p}$$

$$\therefore \boxed{\frac{\partial \delta p}{\partial t} = D_p\frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}}$$

Electron Diffusion Equation

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$$

$$\text{and } J_n(x) = qD_n\frac{dn(x)}{dx} = qD_n\frac{\partial \delta n}{\partial x} \text{ (diffusion)}$$

$$\text{so } \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial}{\partial x}\left(qD_n\frac{\partial \delta n}{\partial x}\right) - \frac{\delta n}{\tau_n}$$

$$\therefore \boxed{\frac{\partial \delta n}{\partial t} = D_n\frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}}$$

Diffusion Equations

Diffusion Length: Steady State

Steady-State Diffusion Equations

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

In steady state, $\frac{\partial \delta p}{\partial t} = 0$

$$\therefore\ D_p \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{\tau_p} \text{ or } \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

In steady state, $\frac{\partial \delta n}{\partial t} = 0$

$$\therefore\ D_n \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{\tau_n} \text{ or } \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

Hole Injection at x=0



(The injection mechanism is not specified.)

L_p is the average distance a hole travels before recombining.
L_n is the average distance an electron travels before recombining.

$$L_n \equiv \sqrt{D_n \tau_n} \text{ and } L_p \equiv \sqrt{D_p \tau_p}$$

Excess Hole Distribution and Current

Solution to Hole Diffusion Equation

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2} \qquad \frac{\partial \delta p}{\partial t} = 0$$

Solution:

$$\delta p(x) = C_1 e^{x/L_p} + C_2 e^{-x/L_p}$$

Boundary Condition:

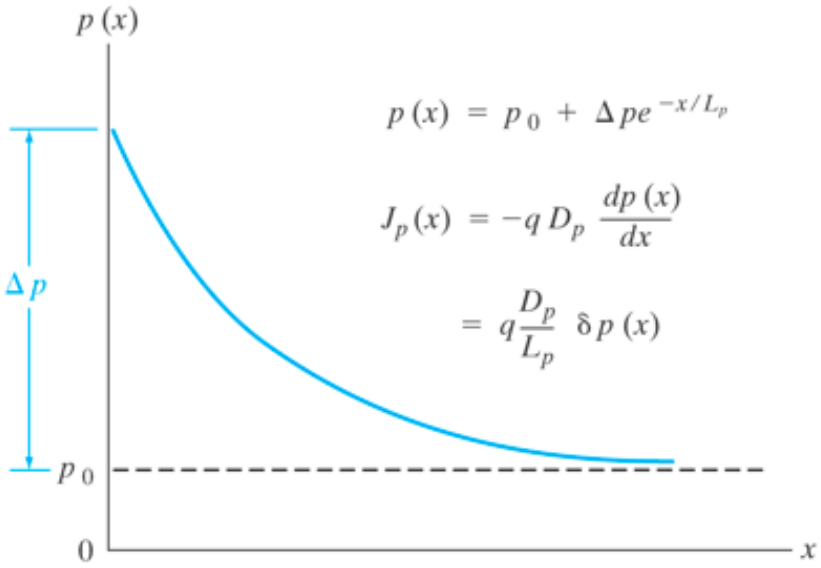
$$\delta p(x=0) = \Delta p = C_1 + C_2$$

$$\delta p(x=\infty) = 0 = C_1 e^{x/L_p}$$

$$C_1 = 0, \quad C_2 = \Delta p$$

$$\delta p(x) = \Delta p \, e^{-x/L_p}$$

Hole Injection at x=0



$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d(p_o + \delta p)}{dx} = -qD_p \frac{d\delta p}{dx}$$

$$= q\frac{D_p}{L_p}\Delta p e^{-x/L_p} = q\frac{D_p}{L_p}\delta p(x)$$

Instructional Objectives (1)

By the time of exam No. 1 (after 17 lectures), the students should be able to do the following:

1. Outline the classification of solids as metals, semiconductors, and insulators and distinguish direct and indirect semiconductors.
2. Determine relative magnitudes of the effective mass of electrons and holes from an E(k) diagram.
3. Calculate the carrier concentration in intrinsic semiconductors.
4. Apply the Fermi-Dirac distribution function to determine the occupation of electron and hole states in a semiconductor.
5. Calculate the electron and hole concentrations if the Fermi level is given; determine the Fermi level in a semiconductor if the carrier concentration is given.
6. Determine the variation of electron and hole mobility in a semiconductor with temperature, impurity concentration, and electrical field.
7. Apply the concept of compensation and space charge neutrality to calculate the electron and hole concentrations in compensated semiconductor samples.
8. Determine the current density and resistivity from given carrier densities and mobilities.
9. Calculate the recombination characteristics and excess carrier concentrations as a function of time for both low level and high level injection conditions in a semiconductor.
10. Use quasi-Fermi levels to calculate the non-equilibrium concentrations of electrons and holes in a semiconductor under uniform photoexcitation.
11. Calculate the drift and diffusion components of electron and hole currents.
12. Calculate the diffusion coefficients from given values of carrier mobility through the Einstein's relationship and determine the built-in field in a non-uniformly doped sample.
13. Calculated diffusion current for steady-state carrier injection

https://my.ece.illinois.edu/courses/description.asp?ECE340